Chapter 3. Common Distributions

Hypergeometric, binomial, multinomial, geometric, negative binomial, Poisson, exponential, gamma, normal

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Conventions

Many distributions below are introduced through a **representation**: $X \overset{d}{=} Z_1 + \cdots + Z_n$ means X has the same distribution as a sum of the indicated variables. A representation makes means, variances and mgfs almost free (Theorem 2.10 and the product rule for mgfs of independent sums). Throughout, $\mathrm{Exp}(\beta)$ and $\mathrm{Gamma}(\alpha, \beta)$ use the *scale* β, so their means are β and $\alpha\beta$.

3.1 The hypergeometric distribution

A population of N units contains D of type "1" and $N-D$ of type "0". The population proportion is $p = D/N$. Draw n units by **simple random sampling**: one at a time, each remaining unit equally likely, without replacement. The number X of type-1 units in the sample has the **hypergeometric distribution** $X \sim H(n; N, D)$:

$$f(x) = \binom{D}{x}\binom{N-D}{n-x}\Big/\binom{N}{n}, \qquad 0 \le x \le D,\ 0 \le n-x \le N-D.$$

Proposition 3.1

$EX = np$ and $\operatorname{Var} X = \dfrac{N-n}{N-1}\,np(1-p)$.

Proof. Using $x\binom{D}{x} = D\binom{D-1}{x-1}$,

$$EX = D\sum_x \binom{D-1}{x-1}\binom{N-1-(D-1)}{n-1-(x-1)}\Big/\binom{N}{n} = D\binom{N-1}{n-1}\Big/\binom{N}{n} = \frac{nD}{N}.$$

The middle sum is Vandermonde's identity, obtained by comparing the coefficient of t^{n-1} on both sides of $(1+t)^{D-1}(1+t)^{N-D} = (1+t)^{N-1}$. In the same way $x(x-1)\binom{D}{x} = D(D-1)\binom{D-2}{x-2}$ gives $E[X(X-1)] = n(n-1)\frac{D(D-1)}{N(N-1)}$. Then

$$\operatorname{Var} X = E[X(X-1)] + EX - (EX)^2 = \frac{N-n}{N-1}\,n\frac{D}{N}\Big(1-\frac{D}{N}\Big).$$

■

Example 3.1 (Acceptance sampling)

A lot of $N = 40$ items contains $D = 4$ defectives. We inspect $n = 5$ at random and accept the lot if at most c defectives are found. Then

$$P(X=0) = \binom{36}{5}\Big/\binom{40}{5} = \frac{376992}{658008} \approx 0.573, \qquad P(X=1) = \frac{4\binom{36}{4}}{\binom{40}{5}} = \frac{235620}{658008} \approx 0.358.$$

So $P(X \le 1) \approx 0.931$ and $P(X \le 2) \approx 0.996$. The rule "accept if $X \le 2$" accepts such a lot more than 99% of the time, so to reject a lot with 10% defectives with probability at least 0.99 we would need a smaller acceptance number or a larger sample.

Binomial approximation. If $N \to \infty$ with $D/N = p$ fixed and n fixed, then

$$\binom{D}{x}\binom{N-D}{n-x}\Big/\binom{N}{n} = \binom{n}{x}\frac{D(D-1)\cdots(D-x+1)\cdot(N-D)\cdots(N-D-n+x+1)}{N(N-1)\cdots(N-n+1)}$$
$$\to \binom{n}{x}p^x(1-p)^{n-x}.$$

Drawing without replacement from a huge population is practically the same as drawing with replacement.

3.2 Binomial and multinomial distributions

If each draw is made *with* replacement, the outcomes $Z_1, \dots, Z_n$ are independent with $P(Z_i = 1) = p$: **Bernoulli trials**, $Z_i \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$. The number of successes is binomial.

Definition (Binomial)

$X \sim B(n,p)$ if $f(x) = \binom{n}{x} p^x (1-p)^{n-x}$, $x = 0, \dots, n$. Representation: $X \overset{d}{=} Z_1 + \cdots + Z_n$, $Z_i \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$.

Proposition 3.2

(a) $EX = np$, $\operatorname{Var} X = np(1-p)$.
(b) $M(t) = (pe^t + q)^n$ with $q = 1 - p$.
(c) If $X_1 \sim B(n_1, p)$ and $X_2 \sim B(n_2, p)$ are independent, then $X_1 + X_2 \sim B(n_1 + n_2, p)$.

Proof. (a) Directly, $EX = \sum x \binom{n}{x} p^x q^{n-x} = np \sum_k \binom{n-1}{k} p^k q^{n-1-k} = np$, and similarly $E[X(X-1)] = n(n-1)p^2$; so $\operatorname{Var} X = n(n-1)p^2 + np - n^2p^2 = np(1-p)$. (Or add the Bernoulli means p and variances $p(1-p)$.) (b) A Bernoulli has mgf $pe^t + q$, and mgfs of independent summands multiply.

(c) The product of the mgfs is $(pe^t + q)^{n_1+n_2}$; uniqueness of mgfs (Theorem 1.7) identifies the distribution. ■

Example 3.2 (Binomial conditioned on its total is hypergeometric)

Let $X_1, \dots, X_N \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$ and $D \le N$. Then the conditional distribution of $X_1 + \cdots + X_D$ given $X_1 + \cdots + X_N = n$ is $H(n; N, D)$. Indeed $X_1 + \cdots + X_D \sim B(D, p)$ and $X_{D+1} + \cdots + X_N \sim B(N - D, p)$ are independent, so

$$P\Big(\sum_{i \le D} X_i = x \,\Big|\, \sum_{i \le N} X_i = n\Big) = \frac{\binom{D}{x} p^x q^{D-x} \binom{N-D}{n-x} p^{n-x} q^{N-D-n+x}}{\binom{N}{n} p^n q^{N-n}} = \frac{\binom{D}{x}\binom{N-D}{n-x}}{\binom{N}{n}}.$$

The parameter p cancels.

Definition (Multinomial)

Each of n independent trials falls in category j with probability p_j, $\sum_j p_j = 1$. The counts $\mathbf{X} = (X_1, \dots, X_k)$ satisfy

$$f(\mathbf{x}) = \frac{n!}{x_1! \cdots x_k!} p_1^{x_1} \cdots p_k^{x_k}, \qquad \sum_j x_j = n,$$

written $\mathbf{X} \sim \text{Multi}(n, \mathbf{p})$. Representation: $\mathbf{X} \overset{d}{=} \mathbf{Z}_1 + \cdots + \mathbf{Z}_n$ with $\mathbf{Z}_i \overset{\text{i.i.d.}}{\sim} \text{Multi}(1, \mathbf{p})$ (one-hot vectors). For $k = 2$, $(X, n - X)$ with $X \sim B(n, p_1)$.

Proposition 3.3

(a) $EX_l = np_l$, $\operatorname{Var} X_l = np_l(1-p_l)$, $\operatorname{Cov}(X_l, X_m) = -np_lp_m$ for $l \neq m$; i.e. $\operatorname{Var}\mathbf{X} = n\{\operatorname{diag}(\mathbf{p}) - \mathbf{pp}'\}$.
(b) $M(\mathbf{t}) = (p_1e^{t_1} + \cdots + p_ke^{t_k})^n$.

Proof. For one trial, $Z_{1l} \in \{0,1\}$ with $P(Z_{1l} = 1) = p_l$, and $Z_{1l}Z_{1m} = 0$ for $l \neq m$. So $\operatorname{Cov}(Z_{1l}, Z_{1m}) = -p_lp_m$, and $M_{\mathbf{Z}_1}(\mathbf{t}) = \sum_l p_le^{t_l}$. Sum over n i.i.d. trials (Theorem 2.11). ■

The negative covariances express the constraint $\sum_j X_j = n$.

3.3 Geometric and negative binomial distributions

In Bernoulli trials with success probability p, the trial number W_1 of the first success satisfies $P(W_1 = x) = q^{x-1}p$, $x = 1, 2, \ldots$: the **geometric distribution** $\operatorname{Geo}(p)$.

Proposition 3.4

(a) $M_{W_1}(t) = pe^t(1-qe^t)^{-1}$ for $t < -\log q$.
(b) $EW_1 = 1/p$, $\operatorname{Var} W_1 = q/p^2$.
(c) (Memorylessness) $P(W_1 > m+k \mid W_1 > m) = P(W_1 > k)$.

Proof. (a) A geometric series: $\sum_x e^{tx}q^{x-1}p = pe^t\sum_x(qe^t)^{x-1}$.

(b) The cgf is $C(t) = t + \log p - \log(1-qe^t)$. Expanding $-\log(1-qe^t)$ in powers of t gives $C(t) = \frac{1}{p}t + \frac{q}{p^2}\frac{t^2}{2} + \cdots$.

(c) $P(W_1 > k) = q^k$. ■

The trial number W_r of the rth success needs $r-1$ successes in the first $x-1$ trials and a success at trial x:

$$P(W_r = x) = \binom{x-1}{r-1}p^rq^{x-r}, \qquad x = r, r+1, \ldots.$$

This is the **negative binomial distribution** $\operatorname{Negbin}(r,p)$. The name comes from the expansion $(1-q)^{-r} = \sum_{k\geq 0}\binom{r+k-1}{k}q^k$; substituting $x = r+k$ shows that the probabilities sum to $p^rp^{-r} = 1$. The waiting times $W_1, W_2 - W_1, \ldots, W_r - W_{r-1}$ are i.i.d. $\operatorname{Geo}(p)$, since

$$P(W_1 = x_1, \ldots, W_r - W_{r-1} = x_r) = \prod_{i=1}^{r} q^{x_i-1}p.$$

Hence the representation $\operatorname{Negbin}(r,p) \stackrel{d}{=} \operatorname{Geo}_1(p) \oplus \cdots \oplus \operatorname{Geo}_r(p)$, where $\oplus$ denotes a sum of independent variables.

Proposition 3.5

For $X \sim \operatorname{Negbin}(r,p)$: $EX = r/p$, $\operatorname{Var} X = rq/p^2$, $M(t) = \{pe^t(1-qe^t)^{-1}\}^r$. Independent $\operatorname{Negbin}(r_i,p)$ variables sum to $\operatorname{Negbin}(\sum r_i, p)$.

Proof. Sum r i.i.d. geometrics (Proposition 3.4); multiply mgfs and use uniqueness. ■

3.4 The Poisson distribution

Motivation. Counts of rare events are modelled by the Poisson distribution, the limit of binomials with np fixed. Its additivity and the link with exponential waiting times make it the basic count model.

Proposition 3.6 (Poisson approximation of the binomial)

If $n \to \infty$ and $np_n \to \lambda > 0$, then $\binom{n}{x} p_n^x (1-p_n)^{n-x} \to e^{-\lambda}\lambda^x/x!$.

Proof.

$$\binom{n}{x} p_n^x (1-p_n)^{n-x} = \frac{n(n-1)\cdots(n-x+1)}{n^x} \cdot \frac{(np_n)^x}{x!} \cdot \Big(1 - \frac{np_n}{n}\Big)^n (1-p_n)^{-x}.$$

The first factor tends to 1, the second to $\lambda^x/x!$, the third to $e^{-\lambda}$ (since $(1+a_n/n)^n \to e^a$ when $a_n \to a$), and the last to 1. ■

$X \sim \text{Poisson}(\lambda)$ if $f(x) = e^{-\lambda}\lambda^x/x!$, $x = 0, 1, \ldots$.

Example 3.3

A batter hits a home run in 2% of at-bats. Over 300 at-bats, $X \sim B(300, 0.02) \approx \text{Poisson}(6)$. Then $P(X = 10) \approx e^{-6}6^{10}/10! \approx 0.041$; the exact binomial value is 0.040.

Proposition 3.7

(a) $M(t) = \exp\{\lambda(e^t - 1)\}$ for all t.
(b) $EX = \text{Var}\, X = \lambda$.
(c) Independent $\text{Poisson}(\lambda_i)$ variables sum to $\text{Poisson}(\sum \lambda_i)$.

Proof. (a) $\sum_x e^{tx} e^{-\lambda}\lambda^x/x! = e^{-\lambda}e^{\lambda e^t}$.

(b) The cgf $\lambda(e^t - 1) = \lambda \sum_{r \geq 1} t^r/r!$ has every cumulant equal to λ.

(c) Multiply mgfs and use uniqueness. ■

The Poisson process. Let N_t count the events in $[0, t]$, with

(a) (stationarity) $N_{s+t} - N_s \overset{d}{=} N_t$ and $N_0 = 0$;
(b) (independent increments) counts on disjoint intervals are independent;
(c) (proportionality) $P(N_h = 1) = \lambda h + o(h)$;
(d) (rareness) $P(N_h \geq 2) = o(h)$ as $h \downarrow 0$.

Theorem 3.8

Under (a)–(d), $N_t \sim \text{Poisson}(\lambda t)$.

Proof. Let $g(x, t) = P(N_t = x)$. Splitting according to the increment on $(t, t+h]$ and using (a), (b), (d),

$$g(x, t+h) = g(x, t)P(N_h = 0) + g(x-1, t)P(N_h = 1) + o(h) = g(x, t)(1 - \lambda h) + g(x-1, t)\lambda h + o(h).$$

Hence $\partial_t g(x,t) = -\lambda g(x,t) + \lambda g(x-1,t)$, i.e. $\frac{d}{dt}\{e^{\lambda t} g(x,t)\} = \lambda e^{\lambda t} g(x-1,t)$, with $g(0,0) = 1$ and $g(x,0) = 0$ for $x \geq 1$. For $x = 0$ this gives $g(0,t) = e^{-\lambda t}$. Inductively, if $g(x-1,t) = e^{-\lambda t}(\lambda t)^{x-1}/(x-1)!$, then $e^{\lambda t} g(x,t) = \int_0^t \lambda(\lambda s)^{x-1}/(x-1)!\, ds = (\lambda t)^x/x!$. ■

Heuristically: split $[0,t]$ into n pieces; each contains an event with probability about $\lambda t/n$, independently, so $N_t \approx B(n, \lambda t/n) \to \text{Poisson}(\lambda t)$.

Example 3.4

A cable has on average 0.04 flaws per meter. In 150 meters the number of flaws is Poisson(6), so $P(\text{at least } 10) = 1 - \sum_{x \leq 9} e^{-6} 6^x/x! \approx 0.084$.

3.5 Exponential and gamma distributions

In a Poisson process with rate λ, the waiting time W_1 for the first event satisfies $\{W_1 > t\} = \{N_t = 0\}$. So $P(W_1 \leq t) = 1 - e^{-\lambda t}$, and W_1 has density $\lambda e^{-\lambda t}$, $t > 0$: the **exponential distribution** $\text{Exp}(1/\lambda)$, with mean $1/\lambda$.

Proposition 3.9

$M_{W_1}(t) = (1 - t/\lambda)^{-1}$ for $t < \lambda$; $EW_1 = 1/\lambda$, $\text{Var}\, W_1 = 1/\lambda^2$; and $P(W_1 > s + t \mid W_1 > s) = P(W_1 > t)$.

Proof. $\int_0^\infty e^{tx} \lambda e^{-\lambda x}\, dx = \lambda/(\lambda - t)$. The cgf $-\log(1 - t/\lambda) = t/\lambda + \frac{1}{2}(t/\lambda)^2 + \cdots$ gives the mean and variance. Memorylessness: $e^{-\lambda(s+t)}/e^{-\lambda s} = e^{-\lambda t}$. ■

For the rth event, $\{W_r > t\} = \{N_t \leq r-1\}$, so $P(W_r \leq t) = 1 - \sum_{k=0}^{r-1} e^{-\lambda t}(\lambda t)^k/k!$. Differentiating, the sum telescopes:

$$f_{W_r}(t) = \lambda e^{-\lambda t}\Big\{ \sum_{k=0}^{r-1} \frac{(\lambda t)^k}{k!} - \sum_{k=1}^{r-1} \frac{(\lambda t)^{k-1}}{(k-1)!} \Big\} = \frac{\lambda^r t^{r-1} e^{-\lambda t}}{(r-1)!}, \qquad t > 0.$$

Allowing any shape $\alpha > 0$ gives the gamma family.

Definition (Gamma)

$X \sim \text{Gamma}(\alpha, \beta)$ if $f(x) = \dfrac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta}$, $x > 0$, where $\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x}\, dx$. Here α is the shape and β the scale, and $\text{Exp}(\beta) = \text{Gamma}(1, \beta)$.

Integration by parts gives $\Gamma(\alpha) = (\alpha - 1)\Gamma(\alpha - 1)$, so $\Gamma(n) = (n-1)!$. Also $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ (substitute $x = z^2/2$ and use $\int \phi = 1$).

Proposition 3.10

(a) $EX = \alpha\beta$, $\text{Var}\, X = \alpha\beta^2$.
(b) $M(t) = (1 - \beta t)^{-\alpha}$ for $t < 1/\beta$.
(c) Independent $\text{Gamma}(\alpha_i, \beta)$ variables with a common scale sum to $\text{Gamma}(\sum \alpha_i, \beta)$. In particular $\text{Gamma}(r, \beta) \overset{d}{=} Z_1 + \cdots + Z_r$ with $Z_i \overset{\text{i.i.d.}}{\sim} \text{Exp}(\beta)$.

Proof. (a) Substituting $y = x/\beta$, $EX^k = \beta^k \Gamma(\alpha + k)/\Gamma(\alpha)$, which gives $\alpha\beta$ and $\alpha(\alpha+1)\beta^2$.

(b) Substitute $y = (1/\beta - t)x$ in $\int e^{tx} f(x)\, dx$.

(c) Multiply the mgfs. ■

In a Poisson process the inter-event times $W_1, W_2 - W_1, \dots$ are i.i.d. $\text{Exp}(1/\lambda)$, and W_r is their sum.

3.6 The normal distribution

Motivation. The normal distribution appears as the limit of sums (central limit theorem) and has unusually convenient algebra: linear combinations of independent normals are normal. Most classical inference is built on it.

Definition (Normal)

$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ is the standard normal density, with cdf Φ. $X \sim N(\mu, \sigma^2)$ has density $\frac{1}{\sigma}\phi(\frac{x-\mu}{\sigma})$.

$\int \phi = 1$: $(\int e^{-z^2/2}\,dz)^2 = \iint e^{-(x^2+y^2)/2}\,dx\,dy = \int_0^{2\pi}\int_0^\infty e^{-r^2/2} r\,dr\,d\theta = 2\pi$.

Proposition 3.11

(a) $EX = \mu$, $\text{Var}\, X = \sigma^2$.
(b) $M(t) = \exp(\mu t + \frac{1}{2}\sigma^2 t^2)$ for all t.
(c) If $X_i \sim N(\mu_i, \sigma_i^2)$ are independent, then $X_1 + X_2 \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$.

Proof. (a) With $z = (x-\mu)/\sigma$: $EX = \mu + \sigma \int z\phi(z)\,dz = \mu$ by symmetry. Also $\int z^2\phi = 2\int_0^\infty z^2\phi(z)\,dz = \frac{2}{\sqrt{\pi}}\Gamma(\frac{3}{2}) = 1$ (substitute $y = z^2/2$), using $\Gamma(\frac{3}{2}) = \frac{1}{2}\sqrt{\pi}$.

(b) $\int e^{tz}\phi(z)\,dz = e^{t^2/2}\int \phi(z-t)\,dz = e^{t^2/2}$ (complete the square). Then $Ee^{t(\sigma Z+\mu)} = e^{\mu t + \sigma^2 t^2/2}$.

(c) Multiply the mgfs. ■

Proposition 3.12 (Representation)

(a) $X \sim N(\mu, \sigma^2) \Rightarrow aX + b \sim N(a\mu + b, a^2\sigma^2)$.
(b) $X \sim N(\mu, \sigma^2) \iff (X-\mu)/\sigma \sim N(0,1) \iff X \overset{d}{=} \sigma Z + \mu$ with $Z \sim N(0,1)$.

Proof. $M_{aX+b}(t) = e^{bt}M_X(at) = \exp\{(a\mu + b)t + \frac{1}{2}a^2\sigma^2 t^2\}$;

(b) is the special case $a = 1/\sigma$, $b = -\mu/\sigma$. ■

So $P(X \le x) = \Phi((x-\mu)/\sigma)$, and $\Phi(-z) = 1 - \Phi(z)$ by symmetry. The upper α point z_α ($P(Z > z_\alpha) = \alpha$) gives the upper α point $\mu + \sigma z_\alpha$ of $N(\mu, \sigma^2)$; e.g. $z_{0.05} = 1.645$, $z_{0.025} = 1.96$.

Example 3.5

$X \sim N(10, 9)$. Then $P(7 < X \le 16) = \Phi(2) - \Phi(-1) = 0.9772 - 0.1587 = 0.8185$ and $P(X > 13) = 1 - \Phi(1) = 0.1587$. The 95th percentile is $10 + 3(1.645) = 14.94$ and the 2.5th percentile is $10 - 3(1.96) = 4.12$.

The normal density arises as the limit of standardized binomial probabilities (de Moivre–Laplace):

$$\sum_{a \le (x-np)/\sqrt{np(1-p)} \le b} \binom{n}{x} p^x (1-p)^{n-x} \to \int_a^b \phi(z)\,dz.$$

This is a special case of the central limit theorem (Chapter 5).

Summary

distribution	pmf/pdf	representation
$B(n,p)$	$\binom{n}{x}p^x(1-p)^{n-x}$	sum of n i.i.d. Bernoulli(p)
Multi($n,\mathbf{p}$)	$\frac{n!}{\prod x_j!}\prod p_j^{x_j}$	sum of n i.i.d. Multi($1,\mathbf{p}$)
Negbin(r,p)	$\binom{x-1}{r-1}p^r(1-p)^{x-r}$	sum of r i.i.d. Geo(p)
Poisson(λ)	$e^{-\lambda}\lambda^x/x!$	limit of $B(n,\lambda/n)$
Gamma(r,β)	$\frac{x^{r-1}e^{-x/\beta}}{\Gamma(r)\beta^r}$	sum of r i.i.d. Exp(β)
$N(\mu,\sigma^2)$	$\frac{1}{\sigma}\phi(\frac{x-\mu}{\sigma})$	$\mu+\sigma Z$, $Z\sim N(0,1)$